

Mini-Project: Build and Evaluate a Practical RAG Assistant

1. Project Overview

In this group mini-project, learners will design, implement, evaluate, and demonstrate a small Retrieval-Augmented Generation (RAG) system.

Throughout the week, each learner will complete individual lab tasks covering the main components of a RAG pipeline. The group project extends and integrates these lab components into one working application that answers questions using a selected knowledge base.

The goal is not to build a production-scale application. The goal is to demonstrate a clear understanding of how documents and other modality data are prepared, retrieved, supplied to a language model, and evaluated.

2. Project Scenario

Your group will build a **RAG Agent** for a selected domain.

The assistant must answer questions using only the information contained in the group's knowledge base. It should retrieve relevant passages, generate a grounded response, and show the sources used. As an extension to the lab, you need to use more than one modality of the data sources.

Possible domains include:

- Organisational policies and employee handbooks
- Course materials or technical documentation
- Product manuals and frequently asked questions
- Public-sector information and service guides
- Sustainability reports
- Healthcare information from approved public documents
- Financial or regulatory guidance
- Research papers on a selected topic

The knowledge base should be small enough to process during the course but sufficiently varied to demonstrate meaningful retrieval.

3. Group Structure

Suggested responsibilities include:

- **Data and ingestion:** document collection, cleaning, chunking, and metadata
- **Retrieval:** embeddings, vector store, search, and retrieval parameters
- **Generation and application:** prompt design, answer generation, citations, and user interface
- **Evaluation and testing:** test dataset, metrics, failure analysis, and improvement

These responsibilities support project organisation, but every learner should understand the complete RAG pipeline.

Groups may reuse and extend individual lab work, but they should integrate it into one coherent system rather than submit several disconnected notebooks.

4. Minimum System Requirements

The completed system must include the following functions.

4.1 Knowledge Base

- Use at least 2 modalities of data such as pdf/images/audio/text/video.
- Record the source or document identity.
- Use content that the group is permitted to process and demonstrate.
- Avoid confidential, personal, copyrighted, or restricted data unless explicit permission has been provided.

4.2 Data Processing

The system must:

- Load the selected data.

- Clean or normalise the extracted content where necessary.
- Split documents into appropriate chunks.
- Attach useful metadata, such as document title, section, page, category, or source.

The group should explain why its chunking strategy is suitable for the selected documents and expected questions.

4.3 Retrieval

The system must:

- Create embeddings for the chunks.
- Store them in a vector index or vector database.
- Convert the user's question into an embedding.
- Retrieve the top relevant chunks.
- Allow the group to configure at least one retrieval parameter, such as top_k, chunk size, overlap, or similarity threshold.

4.4 Grounded Answer Generation

The system must:

- Supply the retrieved evidence to a language model.
- Instruct the model to answer using the supplied context.
- Avoid unsupported answers when the evidence is insufficient.
- Return a clear response to the user.

A suitable fallback could be:

"The available knowledge base does not contain enough information to answer this question."

4.5 Sources and Transparency

Each answer must show at least one of the following:

- Document title or other source title
- Source identifier
- Section or page
- Retrieved text excerpt

The user should be able to identify where the answer came from.

4.6 Evaluation

The group must create a test set containing at least **10 questions**, including:

- Straightforward factual questions
- Questions requiring paraphrase or semantic matching
- Questions for which relevant information appears in more than one document
- At least two questions that cannot be answered from the knowledge base
- At least one difficult or intentionally ambiguous question

For every question, record:

- Expected answer or answer criteria
- Expected source document or relevant passage
- Retrieved evidence
- Generated answer
- Evaluation result
- Short observation or failure explanation

The evaluation must include:

- At least 3 retrieval measure, such as Hit Rate, Recall@k, MRR, or manual relevance judgement
- At least one answer-quality measure, such as correctness, faithfulness, relevance, citation accuracy, or a structured human rubric

- A short analysis of at least three failure cases

5. Required Improvement Experiment

After evaluating the baseline system, the group must implement and test at least one improvement.

Possible improvements include:

- Adjusting chunk size or overlap
- Changing the number of retrieved chunks
- Improving document cleaning
- Adding or improving metadata
- Introducing keyword or hybrid search
- Adding query rewriting
- Revising the RAG prompt
- Removing duplicated context
- Adding a similarity threshold
- Improving abstention behaviour
- Adding reranking

The group must compare the baseline and improved versions using the same test questions.

The comparison should explain:

1. What was changed?
2. What problem was the change intended to solve?
3. Which results improved?
4. Which results became worse or remained unchanged?
5. What trade-offs were introduced?

6. Deliverables

A. Project Report

Recommended length: do not exceed 8 pages, excluding appendices.

The report should contain:

1. Project objective and selected domain
2. Description of the knowledge base
3. RAG architecture
4. Document processing and chunking strategy
5. Embedding and retrieval approach
6. Prompt and answer-generation approach
7. Evaluation dataset and metrics
8. Baseline results
9. Improvement experiment
10. Limitations and recommended next steps
11. Contributions of each group member

B. Group Presentation and Demonstration

Recommended duration: **10 minutes**

Suggested structure:

- Problem and intended users: 1 minute
- Architecture and design decisions: 3 minutes
- Live demonstration: 3 minutes
- Evaluation and improvement results: 2 minutes
- Limitations and lessons learned: 1–2 minutes

Every group member should participate.

7. Assessment Rubric

Criterion	Weight
Technical implementation	40%
Report	20%
Presentation & Demo	40%
Total	100%

Your project should provide an evidence-based answer to this question:

How effectively can your RAG system retrieve the right evidence and produce a grounded, attributable answer for the selected domain?